Chapter 2

Bone as a Living Organ

Hector F. Rios,[1] Jill D. Bashutski,[2] and William V. Giannobile[1,2]

[1] Department of Periodontology and Oral Medicine, University of Michigan, School of Dentistry, Ann Arbor, MI, USA
[2] Department of Biomedical Engineering, College of Engineering, Ann Arbor, MI, USA

Introduction

Bone is a complex organ composed of multiple specialized tissues (osseous, periosteum/endosteum, and bone marrow) that act synergistically and serve multiple functions (Fig. 2-1). Its composition allows the bone tissue to: (1) resist load, (2) protect highly sensitive organs from external forces, and (3) participate as a reservoir of cells and minerals that contribute to systemic homeostasis of the body. Therefore, the concept of "bone as a living organ" integrates the structurally dynamic nature of bone with its capacity to orchestrate multiple mechanical and metabolic functions with important local and systemic implications. Multiple factors exert an effect in this system (e.g. biochemical, hormonal, cellular, biomechanical) and will collectively determine its quality (Ammann & Rizzoli 2003; Marotti & Palumbo 2007; Bonewald & Johnson 2008; Ma *et al*. 2008). The purpose of this chapter is to provide the foundation knowledge of bone development, structure, function, healing, and homeostasis.

Development

During embryogenesis, the skeleton forms by either a direct or indirect ossification process. In the case of the mandible, maxilla, skull, and clavicle, mesenchymal progenitor cells condensate and undergo direct differentiation into osteoblasts, a process known as *intramembranous osteogenesis*.

In contrast, in the mandibular condyle, the long bones and vertebrae form initially through a cartilage template, which serves as an anlage that is gradually replaced by bone. The cartilage-dependent bone formation and growth process is known as *endochondral osteogenesis* (Ranly 2000) (Fig. 2-2).

Intramembranous bone formation

During intramembranous osteogenesis, an ossification center develops through mesenchymal condensation. As the collagen-rich extracellular matrix develops and matures, osteoprogenitor cells undergo further osteoblastic differentiation. On the outer surfaces of the ossification center, a fibrous periosteum forms over a layer of osteoblasts. As new osteoblasts form from the underside of the periosteum, appositional growth occurs. A subpopulation of osteoblasts becomes embedded in the mineralizing matrix and gives rise to the osteocyte lacunocanalicular network. Within the craniofacial complex, most bones develop and grow through this mechanism.

Endochondral bone growth

During endochondral osteogenesis, bones develop through the formation of a cartilaginous template (hyaline cartilage model) that mineralizes and is later

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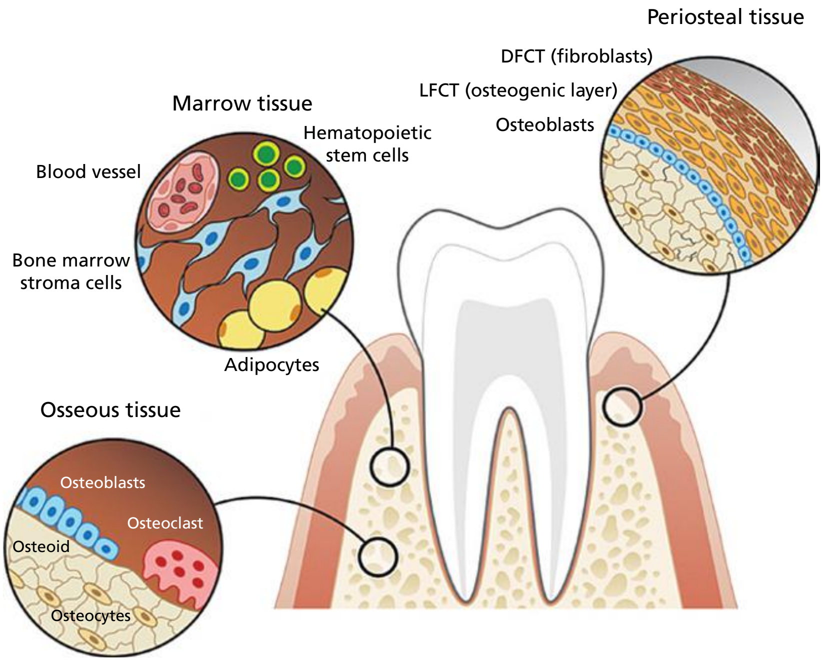


Fig. 2-1 Bone as an organ. The bone organ encompasses a number of complex tissues that synergize during health to execute a number of functions. It serves as a source of stem cells and a reservoir of minerals and other nutrients; it protects a number of delicate organs; and it acts as a mechanosensoring unit that adapts to the environment and individual demands. This figure highlights three main tissues, the cells that are involved in these roles, and the maintenance of the structure and function of bone as an organ. (DFCT, dense fibrous connective tissue; LFCT, loose fibrous connective tissue.)

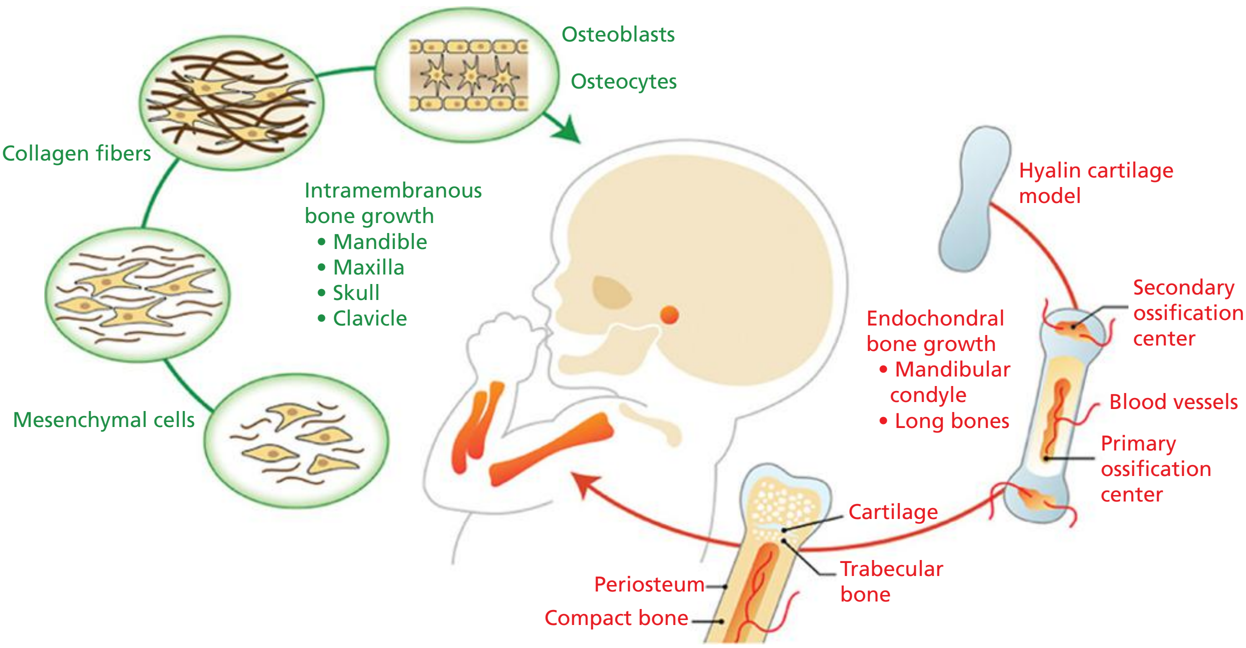


Fig. 2-2 Bone development. There are two types of processes involved in bone development: Intramembranous ossification (green arrow) and endochondral ossification (orange arrow). They primarily differ in the presence of a cartilaginous template during endochondral bone growth. During intramembranous osteogenesis, an ossification center develops through mesenchymal condensation. As the collagen-rich extracellular matrix develops and matures, osteoprogenitor cells undergo further osteoblastic differentiation. A subpopulation of osteoblasts becomes embedded in the mineralizing matrix and gives rise to the osteocyte lacunocanalicular network. Within the craniofacial complex, most bones develop and grow through this mechanism. On the other hand, long bones within the skeleton and the mandibular condyle initially develop through the formation of a cartilaginous template that mineralizes and is later resorbed by osteoclasts and replaced by bone. The endochondral bone growth process leads to the formation of primary and secondary ossification centers that are separated by a cartilaginous structure known as the growth plate. As bone develops and matures through these two processes, structurally distinct areas of compact bone and trabecular bone are formed and maintained through similar bone remodeling mechanisms.

resorbed by osteoclasts and replaced by bone that is laid down afterwards. This process begins during the third month of gestation. The endochondral bone growth process leads to the formation of primary and secondary ossification centers that are separated by a cartilaginous structure known as the growth plate. Following the formation of the primary ossification center, bone formation extends towards both ends of the bone from the center of the shaft. The cartilage cells on the leading edges of ossification die. Osteoblasts cover the cartilagenous trabeculae with woven, spongy bone. Behind the advancing front of ossification, osteoclasts absorb the spongy bone and enlarge the primary marrow cavity. The periosteal collar thickens and extends toward the epiphyses to compensate for the continued hollowing of the primary cavity.

The processes of osteogenesis and resorption occur in all directions. The spaces between the trabeculae become filled with marrow tissue. As the new bone matrix remodels, osteoclasts assist in the formation of primary medullary cavities which rapidly fill with bone marrow hematopoietic tissue. The fibrous, non-mineralized lining of the medullary cavity is the endosteum. Osteoblasts form in the endosteum and begin the formation of endosteal bone. The appositional growth of endosteal bone is closely regulated to prevent closure of the primary marrow cavities and destruction of bone marrow.

Structure

Osseous tissue

Osseous tissue is a specialized connective tissue composed of organic and inorganic elements that mineralizes and is populated by highly specialized cells that regulate its stability (Fig. 2-3a).

Matrix

The organic matrix of bone makes up approximately 30–35% of the total bone weight and is formed of 90% collagen type I and 10% non-collagenous proteins, proteoglycans, glycoproteins, carbohydrates,

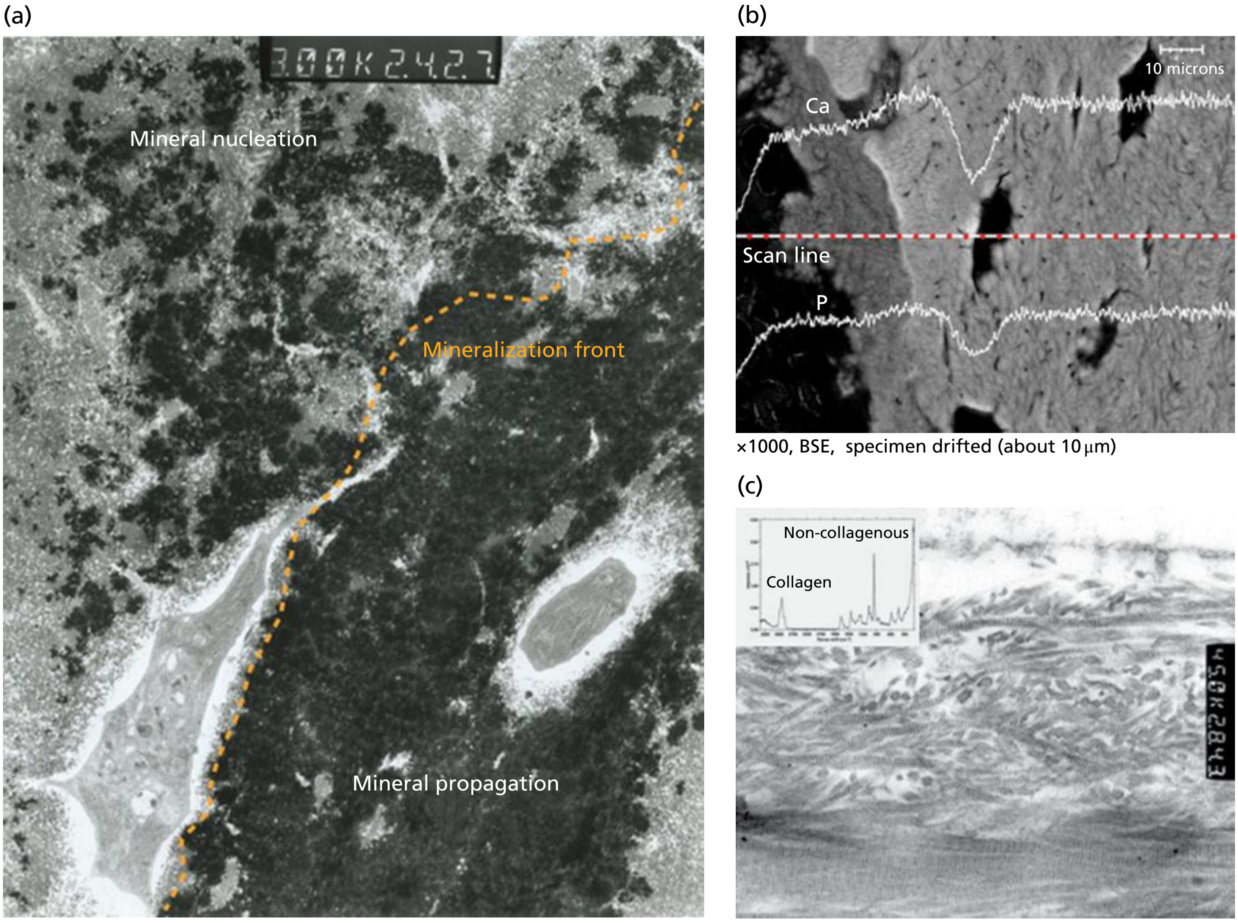


Fig. 2-3 Osseous matrix. The extracellular matrix in bone is particularly abundant as compared to its cellular counterpart. (a) Osseous matrix has the unique ability to mineralize: a process that requires the support of organic components and the assistance of highly specialized cells. (b) Calcium and phosphorus are present in the form of hydroxyapatite crystals. These crystals tend to follow the organic scaffold in the bone matrix. The orange dashed line represents a linear scan that emphasizes the high content of calcium and phosphorus in the mature bone, as shown by the energy dispersive X-ray spectroscopy analysis. (c) Collagen fibers as well as non-collangenous proteins are abundant in the matrix and are often found to be arranged in a preferential direction, as shown by the Raman spectroscopy.

and lipids. The organic matrix is synthesized by osteoblasts, and while it is still unmineralized, is known as osteoid. Within the collagen fibers, mineral nucleation occurs as calcium and phosphate ions are laid down and ultimately form hydroxyapatite crystals. Non-collagenous proteins along the surface of the collagen fibers assist in the propagation of the mineral and the complete mineralization of the matrix.

Inorganic components

Hydrated calcium and phosphate in the form of hydroxyapatite crystals [$3Ca_3(PO_4)_2(OH)_2$] are the principal inorganic constituent of the osseous matrix. Mineralization is clearly depicted in backscatter scanning electron imagines as a bright signal (Fig. 2-3b). Different degrees of mineralization are noticeable within the mature bone. Specific elements within the mineral can be further identified by energy-dispersive X-ray spectroscopy (EDS). In Fig. 2-3b, characteristic peaks of calcium and phosphorus are significantly pronounced in bone, as expected from their high content within the hydroxyapatite crystals.

Organic components

Bone is initially laid down as a purely organic matrix rich in collagen as well as in other non-collagenous molecules (Fig. 2-3c). Chemical analysis of bone by Raman spectroscopy clearly highlights this organic counterpart in the matrix. The transition from a purely organic matrix to a mineralized matrix is clearly depicted in the transmission electron micrograph in Fig. 2-3a as an osteocyte becomes embedded within the mineralized mature matrix. As the matrix matures, mineral nucleation and propagation is mediated by the organic components in the extracellular matrix. Figure 2-3a shows the aggregation of mineral crystals, forming circular structures. As the mineral propagates along the collagen fibrils, a clear mineralization front forms and clearly demarcates the transition between the osteoid area and the mature bone.

Mineralization

The initiation of the mineral nucleation within osteoid typically occurs within a few days of the laying down of calcium and phosphate ions, but maturation is completed through the propagation of the hydroxyapatite crystals over several months and as new matrix is synthesized (Fig. 2-3a). In addition to providing the bone with its strength and rigidity to resist load and protect highly sensitive organs, the mineralization of the osteoid allows the storage of minerals that contribute to systemic homeostasis.

Cells

Within bone, different cellular components can be identified. The distinct cell populations include osteogenic precursor cells, osteoblasts, osteoclasts, osteocytes, and hematopoietic elements of bone marrow. This chapter will focus on the three main functional cells that are ultimately responsible for the proper skeletal homeostasis.

Osteoblasts (Fig. 2-4)

Osteoblasts are the primary cells responsible for the formation of bone; they synthesize the organic extracellular matrix (ECM) components and control the mineralization of the matrix (Fig. 2-4a, b). Osteoblasts are located on bone surfaces exhibiting active matrix deposition and may eventually differentiate into two different types of cells: bone lining cells and osteocytes. Bone lining cells are elongated cells that cover a surface of bone tissue and exhibit no synthetic activity. The osteoblasts are fully differentiated cells and lack the capacity for migration and proliferation. Thus, for bone formation at a given site, undifferentiated mesenchymal progenitor cells, driven by the expression of a gene known as Indian hedgehog (*Ihh*) and later by *RUNX2*, and osteoprogenitor cells migrate to the site and proliferate to become osteoblasts (Fig. 2-4c). The determined osteoprogenitor cells are present in the bone marrow, in the endosteum, and in the periosteum that covers the bone surface. Such cells possess an intrinsic capacity to proliferate and differentiate into osteoblasts. The differentiation and development of osteoblasts from osteoprogenitor cells are dependent on the release of osteoinductive or osteopromotive growth factors (GFs), such as bone morphogenetic proteins (BMPs), and other growth factors, such as insulin-like growth factor (IGF), platelet-derived growth factor (PDGF), and fibroblast growth factor (FGF).

Osteocytes (Fig. 2-5)

Osteocytes are stellate-shaped cells that are embedded within the mineralized bone matrix in spaces known as lacunae (Fig. 2-5a, b). They maintain a network of cytoplasmic processes known as dendrites (Fig. 2-5c). These osteocyte cytoplasmic projections extend through cylindrical encased compartments commonly referred to as canaliculi (Bonewald 2007). They extend to different areas and contact blood vessels and other osteocytes (Fig. 2-5d, e). The osteocyte network is therefore an extracellular and intracellular communication channel that is sensitive at the membrane level to shear stress caused by the flow of fluid within the canaliculi space as the result of mechanical stimuli and bone deformation. Osteocytes translate mechanical signals into biochemical mediators that will assist with the orchestration of anabolic and catabolic events within bone. This arrangement allows osteocytes to (1) participate in the regulation of blood calcium homeostasis and (2) sense mechanical loading and transmit this information to other cells within the bone to further orchestrate osteoblast and osteoclast function (Burger *et al*. 1995; Marotti 2000). Different bone diseases and disorders affect the arrangement of the osteocyte lacunocanalicular system, causing significant alteration to this important cellular network (Fig. 2-6).

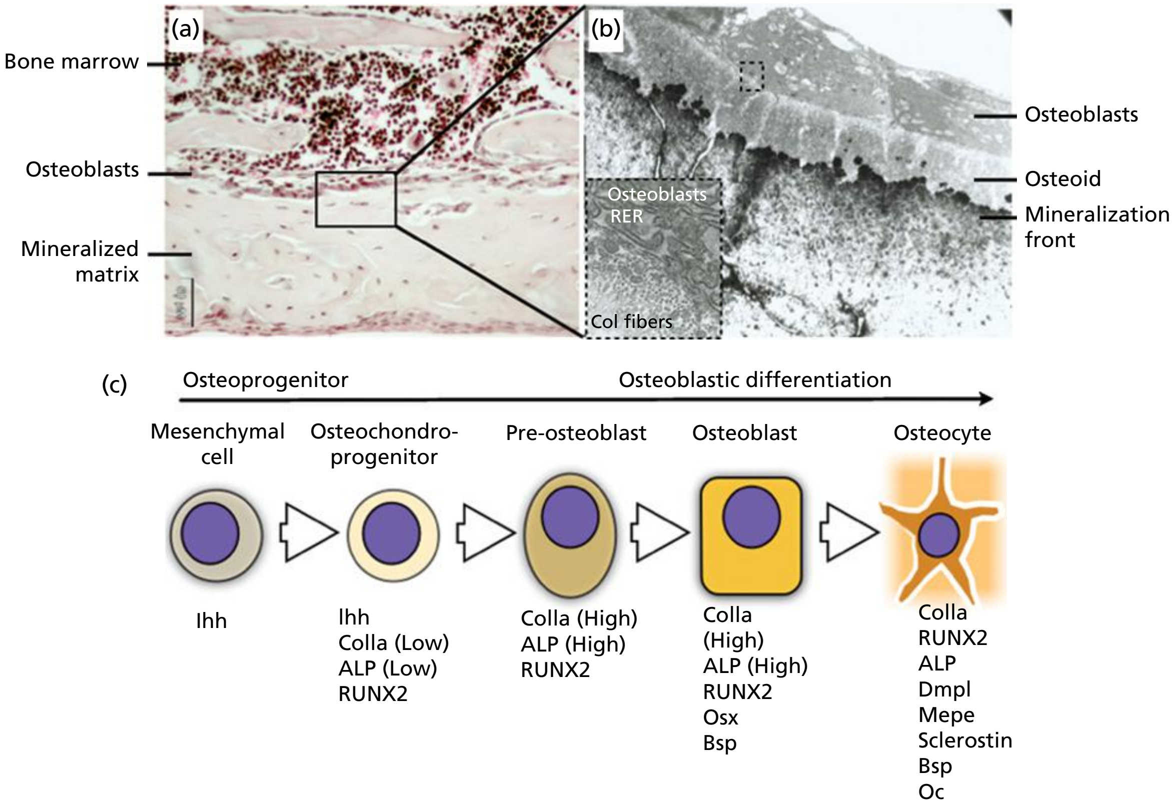


Fig. 2-4 Osteoblast. Osteoblasts are derived from bone marrow osteoprogenitor cells and are responsible for the synthesis of the immature bone matrix known as osteoid. (a) Group of osteoblasts that line the mature bone that contains cells embedded within the mineralized matrix. (b) Further detail of the osteoblasts lining the mature bone is clearly visualized with transmission electron microscopy (TEM). The abundant rough endoplasmic reticulum (RER) and Golgi apparatus within these cells reflect their high metabolic activity. (c) The key molecules involved in the differentiation of an osteoprogenitor cell through to a mature terminally differentiated osteocyte.

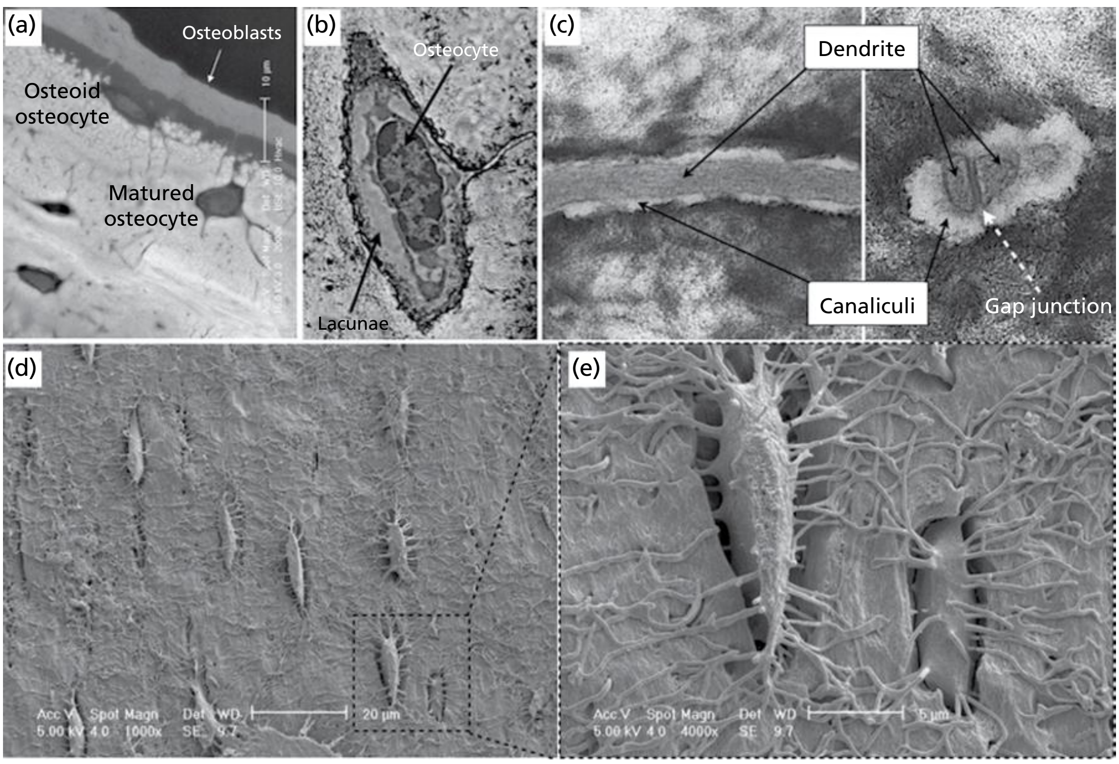


Fig. 2-5 Osteocytes. The osteocyte can be defined as the orchestrator of the remodeling process within bone. (a) As bone matrix is synthesized, a number of osteoblasts become embedded within the osteoid, which later mineralizes and resides in the mature matrix as osteocytes, as shown in this backscatter scanning electron micrograph (SEM) treated with osmium to allow the visualization of the cell. (b) Osteocytes reside within a well-defined space in bone known as the osteocyte lacuna. (c) Transmission electron micrograph of a dendrite within a canaliculi, showing the space through which fluid flows; the shear stress from this stimulates the surface of the osteocyte cell membrane. This unique biologic architectural characteristic of the osteocyte and the lacunocanalicular network represent the foundation that allows the conversion of mechanical stimuli into the biochemical signals necessary for bone homeostasis. (d and e) SEM of a casted lacunocanalicular network allows the visualization of the degree of connectivity between osteocytes and the regularly arranged canalicular structures.

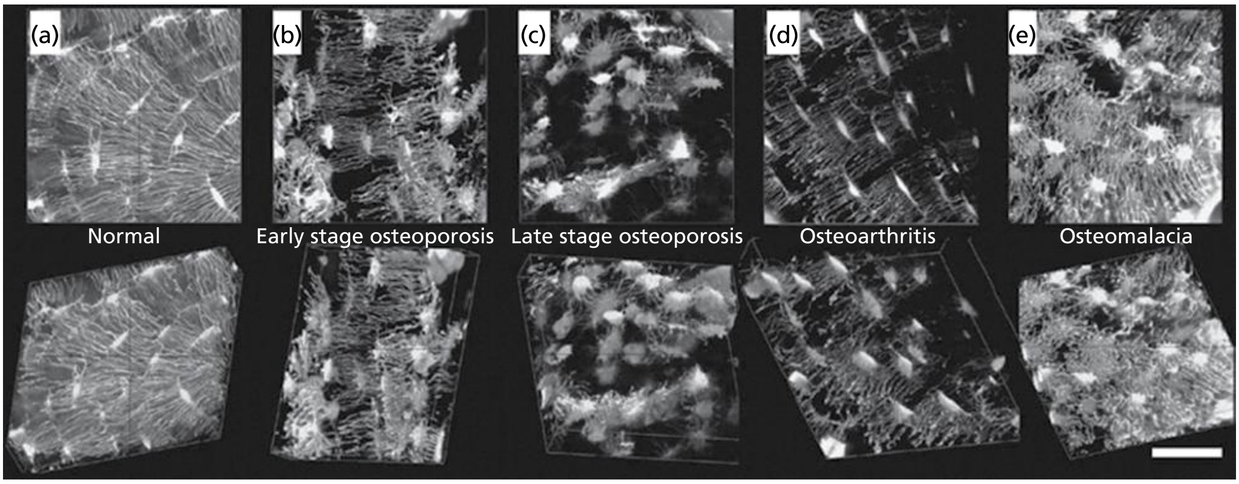


Fig. 2-6 Osteocytes: Lacunocanalicular system in disease. (a) In healthy bone, a high density osteocyte system is established throughout the mature matrix and is characterized by high cellular interconnectivity. With disease, the system is significantly disrupted, leading to important functional alterations. (b, c) In osteoporosis, osteocytic density changes and an apparent decrease in cellular interconnectivity is observed. (d) In osteoarthritis, the canalicular system is altered, but with no major lacunar changes. (e) In osteomalacia, the entire osteocyte lacunocanalicular system appears disrupted due to the altered mineralization pattern. (Source: Knothe Tate *et al*. 2004. Reproduced with permission from Elsevier.)

Osteoclasts

The bone formation activity is consistently coupled to bone resorption that is initiated and maintained by osteoclasts. These cells have the capacity to develop and adhere to bone matrix and then to secrete acid and lytic enzymes that degrade and break down the mineral and organic components of bone and calcified cartilage (Fig. 2-7a–c). The matrix degradation process results in the formation of a specialized extracellular compartment known as *Howship's lacuna* (Rodan 1992; Vaananen & Laitala-Leinonen 2008). Osteoclasts are specialized multinucleated cells that originate from the monocyte/macrophage hematopoietic lineage. The differentiation process is driven initially by the expression of the transcription factor PU-1. Macrophage colony-stimulating factor (M-CSF) engages osteoclasts in the differentiation pathway and promotes their proliferation and expression of RANKL. At this stage, RANKL-expressing stromal cells interact with preosteoclasts and further commit them to differentiation along the osteoclast lineage (Figs. 2-7d, 2-8).

Periosteal tissue

The periosteum is a fibrous sheath that lines the outer surface a long bone's shaft (diaphysis), but not the articulating surfaces. Endosteum lines the inner surface of all bones. The periosteum consists of dense irregular connective tissue. The periosteum is divided into a dense, fibrous, vascular layer (the "fibrous layer") and an inner, more loosely arranged, connective tissue inner layer (the "osteogenic layer") (see Fig. 2-1). The fibrous layer is mainly formed of fibroblasts, while the inner layer contains osteoprogenitor cells.

Osteoblasts derived from the "osteogenic layer" are responsible for increasing the width of long bones and the overall size of the other bone types. In the context of a fracture, progenitor cells from the periosteum differentiate into osteoblasts and chondroblasts, which are essential in the process of stabilizing the wound.

In contrast to the osseous tissue, the periosteum has nociceptive nerve endings, making it very sensitive to manipulation. It also allows the passage of lymphatics and blood vessels into and out of bone, providing nourishment. The periosteum anchors tendons and ligaments to bone by strong collagenous fibers in the "osteogenic layer", called Sharpey's fibers, which extend to the outer circumferential and interstitial lamellae. It also provides an attachment for muscles and tendons.

Bone marrow

The bone marrow consists of hematopoietic tissue islands, stromal cells, and adipose cells surrounded by vascular sinuses interspersed within a meshwork of trabecular bone (see Fig. 2-1). The bone marrow is the major hematopoietic organ, a primary lymphoid tissue (responsible for the production of erythrocytes, granulocytes, monocytes, lymphocytes, and platelets) and an important source of stem cells.

Types

There are two types of bone marrow: red marrow, which consists mainly of hematopoetic tissue, and yellow marrow, which is mainly made up of adipocytes. Erythrocytes, leukocytes, and platelets arise in red marrow. Both types of bone marrow contain numerous blood vessels and capillaries. At birth, all bone marrow is red. With age, more and more of it is converted to the yellow type; only around half of adult bone marrow is red. In cases of severe blood loss, the body can convert yellow marrow back to red marrow to increase blood cell production.

Cells

The stroma of the bone marrow is not directly involved in the primary function of hematopoiesis. However, it serves an indirect role by indirectly providing the

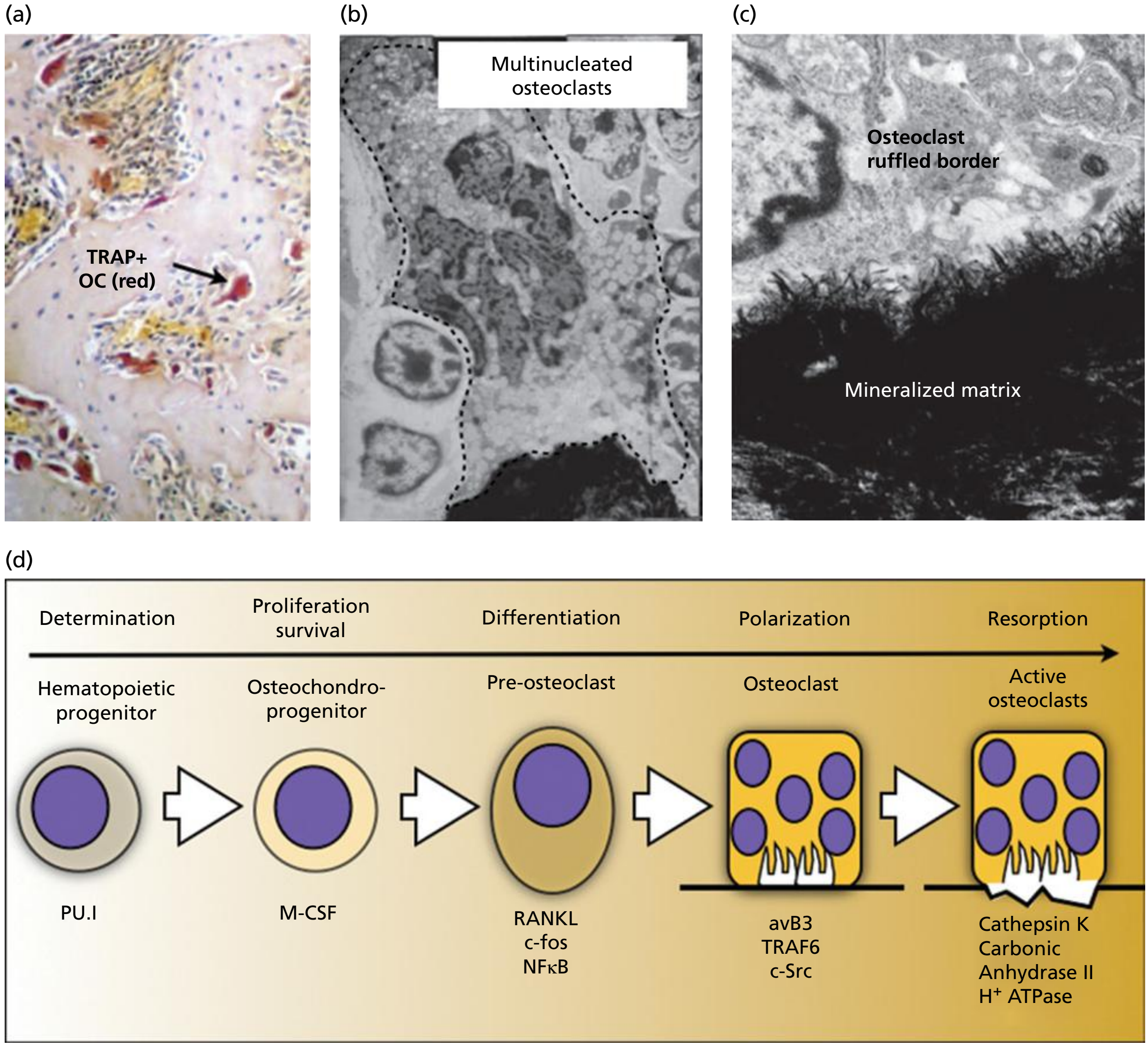


Fig. 2-7 Osteoclasts. (a) Histologically, osteoclasts can be depicted morphologically as multinucleated cells attached to bone matrix using special staining such as with the tartrate resistant acid phosphatase (TRAP) stain (arrow). (OC, osteoclast). (b) Transmission electron micrograph of a multinucleated osteoclast attached to the mineralized bone matrix is delineated by the dotted line. (c) Ruffled border at the resorbing end of the cells. (d) Osteoclasts are derived from cells of the macrophage/monocyte lineage and represent the bone resorbing units within the skeleton. The key molecules involved in the early events of differentiation of a hematopoietic progenitor through to a mature functional osteoclast are shown.

ideal hematopoietic microenvironment. For instance, it generates colony stimulating factors, which have a significant effect on hematopoiesis. Cells that constitute the bone marrow stroma are:

- Fibroblasts
- Macrophages
- Adipocytes
- Osteoblasts
- Osteoclasts
- Endothelial cells.

Stem cells

The bone marrow stroma contains mesenchymal stem cells (MSCs), also called *marrow stromal cells*. These are multipotent stem cells that can differentiate into a variety of cell types. MSCs have been shown to differentiate, *in vitro* or *in vivo*, into osteoblasts, chondrocytes, myocytes, adipocytes, and beta-pancreatic

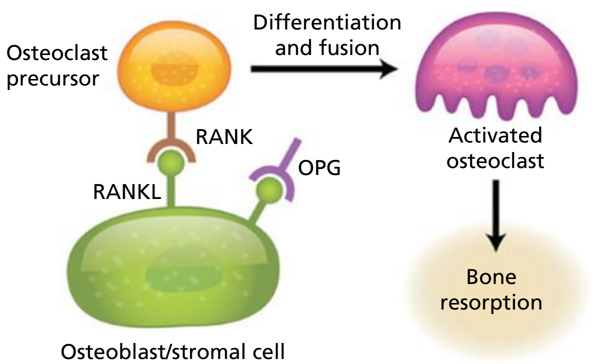


Fig. 2-8 Bone formation/resorption coupling. Bone formation and resorption processes are intimately linked. The osteoblastic/stromal cells provide an osteoclastogenic microenvironment by presenting RANKL to the osteoclast precursor, triggering their further differentiation and fusion, and leading to the formation of multinucleated and active osteoclasts. This process is modulated by inhibitors of these interactions such as osteoprotegerin (OPG). In addition, bone formation by osteoblasts depends on the preceding resorption by osteoclasts.

islet cells. MSCs can also transdifferentiate into neuronal cells. In addition, the bone marrow contains hematopoietic stem cells, which give rise to the three classes of blood cells that are found in the circulation: leukocytes, erythrocytes, and platelets.

Function

The main functions of bone are to provide locomotion, organ protection, and mineral homeostasis. Mechanical tension, local environment factors, and systemic hormones influence the balance between bone resorption and deposition. The distinct mechanical properties of bone contribute to its strength and ability to allow movement. In addition, an intricate series of interactions between cells, matrix, and signaling molecules maintains calcium and phosphorus homeostasis within the body, which also contributes to mechanical strength.

Mechanical properties

Bone is a highly dynamic tissue that has the capacity to adapt based on physiologic needs. Hence, bone adjusts its mechanical properties according to metabolic and mechanical requirements (Burr *et al*. 1985; Lerner 2006). As mentioned earlier, calcium and phosphorus comprise the main mineral components of bone in the form of calcium hydroxyapatite crystals. Hydroxyapatite regulates both the elasticity, stiffness, and tensile strength of bone. The skeletal adaptation mechanism is primarily executed by the processes of bone resorption and bone formation, referred to collectively as bone remodeling (Fig. 2-9). Bone is resorbed by osteoclasts, after which new bone is deposited by osteoblasts (Raisz 2005). From the perspective of bone remodeling, it has been proposed that osteoclasts recognize and are attracted to skeletal sites of compromised mechanical integrity, and initiate the bone remodeling process for the purpose of inducing the generation of new bone that is mechanically competent (Parfitt 1995, 2002).

In general, bone tissue responds to patterns of loading by increasing matrix synthesis, and altering composition, organization, and mechanical properties (Hadjidakis & Androulakis 2006). Evidence indicates that the same holds true for bone under repair. When bone experiences mechanical loading, osteoclast mechanoreceptors are directly stimulated, which begins the bone turnover process to regenerate and repair bone in the area. In addition, pressure increases M-CSF expression, increasing osteoclast differentiation in the bone marrow (Schepetkin 1997). Osteoclasts are also indirectly stimulated through osteoblasts and chondrocytes secreting prostaglandins in response to mechanical pressure. The extracellular matrix can also promote bone turnover through signaling. Mechanical deformation of the matrix induces electric potentials that stimulate osteoclastic resorption.

Bone strength is determined by a combination of bone quality, quantity, and turnover rate. It is well established that a loss of bone density, or quantity, decreases bone strength and results in increased fracture incidence. However, several pathologic conditions characterized by increased bone density, such as Paget's disease, are also associated with decreased bone strength and increased fracture incidence, so quality of bone is also an important factor in determining bone strength.

Metabolic properties

Calcium homeostasis is of major importance for many physiologic processes that maintain health (Bonewald 2002; Harkness & Bonny 2005). Osteoblasts deposit calcium by mechanisms including phosphate and calcium transport with alkalinization to absorb acid produced by mineral deposition; cartilage calcium mineralization occurs by passive diffusion

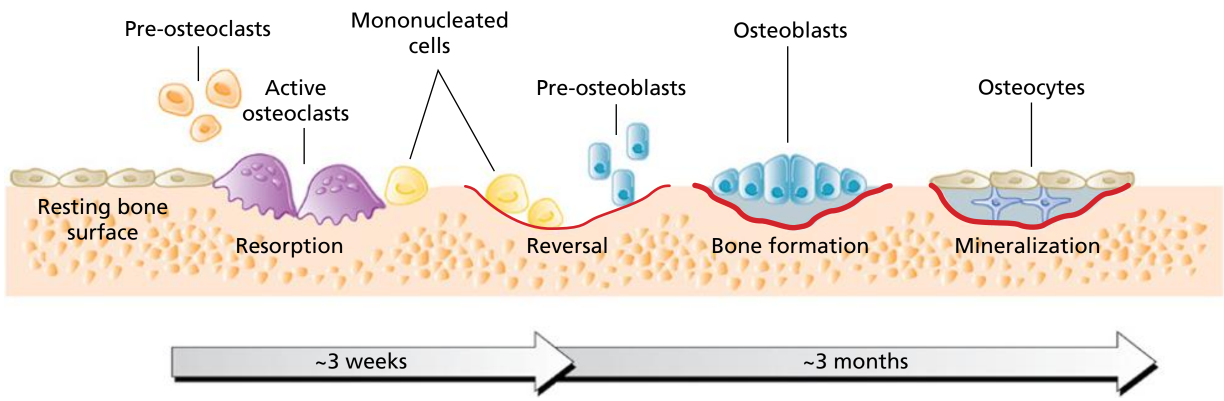


Fig. 2-9 Bone remodeling. The bone remodeling cycle involves a complex series of sequential steps that are highly regulated. The "activation" phase of remodeling is dependent on the effects of local and systemic factors on mesenchymal cells of the osteoblast lineage. These cells interact with hematopoietic precursors to form osteoclasts in the "resorption" phase. Subsequently, there is a "reversal" phase during which mononuclear cells are present on the bone surface. They may complete the resorption process and produce the signals that initiate bone formation. Finally, successive waves of mesenchymal cells differentiate into functional osteoblasts, which lay down matrix in the "formation" phase. (Source: McCauley & Nohutcu 2002. Reproduced from the American Academy of Periodontology.)

and phosphate production. Calcium mobilization by osteoclasts is mediated by acid secretion. Both bone-forming and bone-resorbing cells use calcium signals as regulators of differentiation and activity (Sims & Gooi 2008). This has been studied in more detail in osteoclasts: both osteoclast differentiation and motility are regulated by calcium.

Calcium is obtained from the diet even though bone is the major store of calcium and a key regulatory organ for calcium homeostasis. Bone, in major part, responds to calcium-dependent signals from the parathyroid glands and via vitamin D metabolites, although it responds directly to extracellular calcium if parathyroid regulation is lost. Serum calcium homeostasis is achieved through a complex regulatory process whereby a balance between bone resorption, absorption, and secretion in the intestine, and reabsorption and excretion by the kidneys is tightly regulated by osteotropic hormones (Schepetkin 1997). The balance of serum ionized calcium blood concentrations results from a complex interaction between parathyroid hormone (PTH), vitamin D, and calcitonin. Other osteotropic endocrine hormones that influence bone metabolism include thyroid hormones, sex hormones, and retinoic acids. In addition, fibroblast growth factor aids in phosphate homeostasis. Figure 2-10 reflects how input from the diet and from the bones and excretion via the gastrointestinal tract and urine maintain homeostasis.

Vitamin D is involved in the absorption of calcium, while PTH stimulates calcium release from the bone, reduces its excretion from the kidney, and assists in the conversion of vitamin D into its biologically active form (1,25-dihydroxycholecalciferol) (Holick 2007). Decreased intake of calcium and vitamin D and estrogen deficiency may also contribute to calcium deficiency (Lips *et al.* 2006). Hormonal factors such as retinoids, thyroid and steroid hormones are capable of passing through biologic membranes and interacting with intracellular receptors to have a major impact on the rate of bone resorption. Lack of estrogen increases bone resorption as well as decreases the formation of new bone (Harkness & Bonny 2005). Osteocyte apoptosis has also been documented in estrogen deficiency. In addition to estrogen, calcium metabolism plays a significant role in bone turnover, and deficiency of calcium and vitamin D leads to impaired bone deposition.

Circulating PTH regulates serum calcium and is released in conditions of hypocalcemia. PTH binds to osteoblast receptors, increasing the expression of RANKL and the binding of RANKL to RANK on osteoclasts (McCauley & Nohutcu 2002). This signaling stimulates bone remodeling by activating osteoclasts with the final goal of promoting calcium

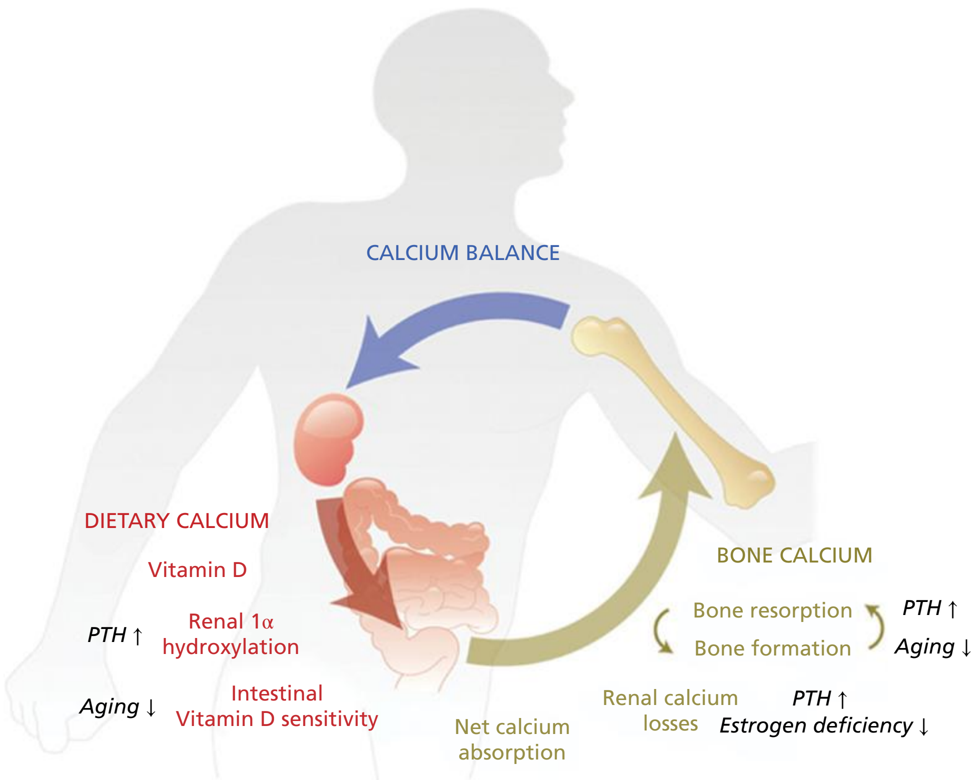


Fig. 2-10 Calcium and bone metabolism. Calcium homeostasis is of major importance for many physiologic processes that maintain health. The balance of serum ionized calcium blood concentrations results from a complex interaction between parathyroid hormone (PTH), vitamin D, and calcitonin. The figure reflects how input from the diet and from the bones, and excretion via the gastrointestinal tract and urine, maintain homeostasis. Vitamin D is involved in the absorption of calcium, while PTH stimulates calcium release from the bone, reduces its excretion from the kidney, and assists in the conversion of vitamin D into its biologically active form (1,25-dihydroxycholecalciferol). Decreased intake of calcium and vitamin D, and estrogen deficiency may also contribute to calcium deficiency.

release from bone. A secondary function of PTH is to increase calcium reabsorption from the kidney. When administered therapeutically at low, intermittent doses, PTH can act as an anabolic agent to promote bone formation, although the mechanism of this action is not well understood.

T cells produce calcitonin, a 32-amino acid peptide whose main physiologic role is the suppression of bone resorption. Calcitonin receptors are present in high numbers on osteoclasts and their precursors (Schepetkin 1997). Thus, calcitonin is able to act directly on osteoclast cells at all stages of their development to reduce bone resorption through preventing fusion of mononuclear preosteoclasts, inhibiting differentiation, and preventing resorption by mature osteoclasts (McCauley & Nohutcu 2002). The concentration and phosphorylation of calcitonin receptors decreases in the presence of calcitonin. As a result, the effect of calcitonin on osteoclasts is transient and thus is not used for clinical therapeutic applications.

Skeletal homeostasis

Healing

Healing of an injured tissue usually leads to the formation of a tissue that differs in morphology or function from the original tissue. This type of healing is termed *repair*. Tissue *regeneration*, on the other hand, is a term used to describe a healing that leads to complete restoration of morphology and function. The healing of bone tissue includes both regeneration and repair phenomena, depending on the nature of the injury.

Repair

Trauma to bone tissue, whether repeated stress or a single, traumatic episode, most commonly results in fracture. When bone is damaged, a complex and multistage healing process is immediately initiated in order to facilitate repair. Tissue and cell proliferation are mediated at different stages by carious growth factors, inflammatory cytokines, and signaling molecules. Although it is a continuous process, bone repair can be roughly divided into three phases: inflammation, reparative, and remodeling (Hadjidakis & Androulakis 2006).

The inflammation phase begins immediately after tissue injury and lasts for approximately 2 weeks (Fazzalari 2011). The initial step in the repair process is the formation of a blood clot. Cytokine release from injured cells then recruits inflammatory cells into the area, where macrophages begin phagocytosis of damaged tissue and cells. Osteoclasts begin the process of resorbing damaged bone in the area to recycle mineral components. In addition, cells from myeloid and mesenchymal cell lineages are recruited to the area where they begin to differentiate into osteoblasts and chondroblasts. At this point, the RANKL-to-osteoprotegerin (OPG) ratio is reduced.

The reparative phase is characterized by the formation of a soft callous where new bone matrix and cartilage scaffolding begins to form. Osteoblasts and chondroblasts produce a protein scaffold to create this callus, which is slowly mineralized to form a hard callus. The hard callus is composed of immature woven bone. The initiation of cartilage and periosteal woven bone formation is primarily mediated through early up-regulation of interleukin 6 (IL-6), OPG, vascular endothelial growth factor (VEGF), and BMPs (Fazzalari 2011). The process of soft to hard callus formation occurs approximately 6–12 weeks from the time of bone fracture.

In the final stage of repair, known as the remodeling phase, the bone matrix and cartilage are remodeled into mature bone. Woven bone is eventually converted into mature lamellar bone through normal bone turnover mediated by osteoblast–osteoclast coupling. Adequate vitamin D and calcium are critical for proper bone repair and their levels may, in part, dictate the rate of repair. The time for the remodeling stage varies depending upon individual bone metabolism, but usually require months from the time of injury.

Regeneration

Ideal bone healing promotes tissue formation in such a way that the original structure and function is preserved. This is in contrast to tissue repair, which merely replaces lost tissue with immature tissue and does not completely restore function.

Over time, bone sustains damage from mechanical strain, overloading, and other forms of tissue injury that results in microfractures and other defects in the bony architecture. In order to prevent greater injury, the bone undergoes a natural remodeling process to regenerate or renew itself. The turnover rates of individual bones is unique, although the average turnover rate is 10% (McCauley & Nohutcu 2002).

Regeneration of bone tissue involves the coupling of bone formation and resorption in a basic multicellular unit (BMU) (Sims & Gooi 2008) (Fig. 2-11). In this process, bone resorption by osteoclasts occurs first over a period of 3–4 weeks, along with cellular signaling to promote osteoblast recruitment to the area. Osteoblasts then form bone for a period of 3–4 months, with a quiescent period between bone resorption and formation, called the reversal phase. Trabecular bone undergoes a significantly higher degree of bone turnover than cortical bone (McCauley & Nohutcu 2002). In a rodent alveolar bone healing model, this process occurs more rapidly, allowing appreciation of the cellular and molecular events that occur during the maturation of the newly regenerated bone (Figs. 2-12, 2-13) (Lin *et al*. 2011).

Bone regeneration is a normal process, but in some cases there is a need to regenerate bone at an increased rate or to overcome the effects of pathologic bone disorders. Therapeutic strategies to promote bone

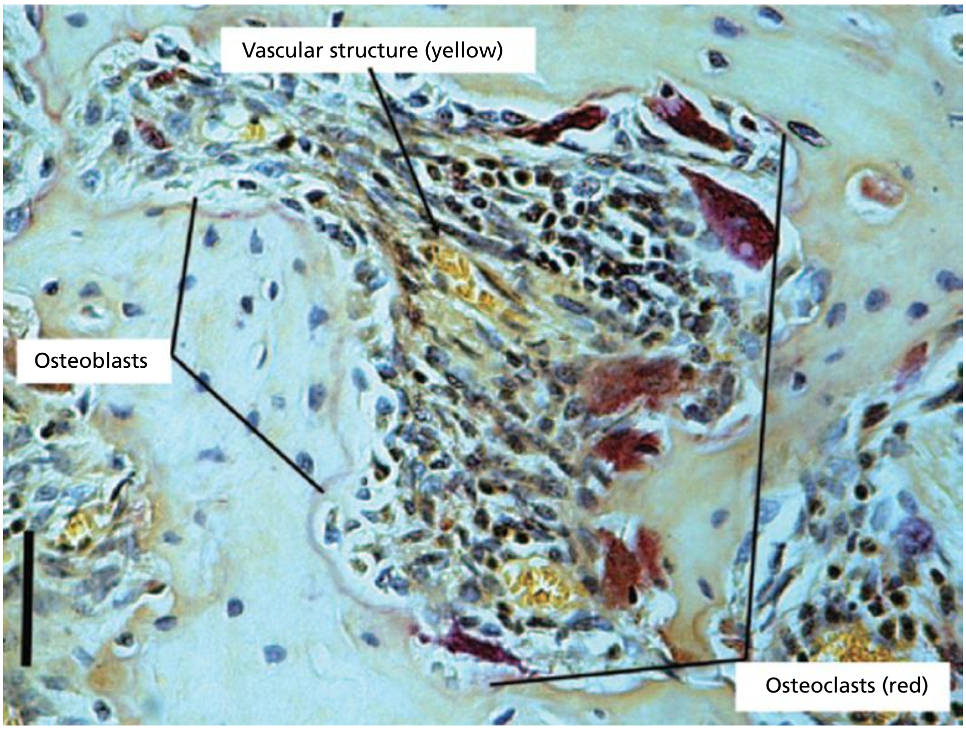


Fig. 2-11 Bone multicellular units (BMUs). Bone remodeling occurs in local groups of osteoblasts and osteoclasts called BMUs; each unit is organized into an osteoclast reabsorbing front, followed by a trail of osteoblasts reforming the bone to fill the defect left by osteoclasts. The red staining (tartrate acid phosphatase) highlights the resorption front. Note the increased number of multinucleated osteoclasts in this area.

regeneration include bone grafting from various sources, epithelial–occlusive barrier membranes, antiresorptive agents, anabolic agents, and growth factors to promote osteoblast differentiation and proliferation.

When alterations in bone turnover occur, skeletal homeostasis is disrupted, resulting in conditions of increased or decreased bone mineral density (BMD), or bone necrosis, and often accompanied by a decrease in bone strength. A wide variety of conditions can alter bone homeostasis and these include cancer, menopause, medications, genetic conditions, nutritional deficiencies, or infection. Some of these etiologies, such as vitamin D deficiency, are easily treatable, whereas others, such as genetic mutations, are typically treated through managing symptoms. Alterations in bone homeostasis cause a wide array of symptoms, including increased fracture incidence, bone pain, and other skeletal deformities that result in a high degree of morbidity and in some cases mortality. A brief review of the more common conditions is given below.

Disorders

Osteoporosis

Osteoporosis is a common condition characterized by both alterations in the macro- and micro-architecture of the bone (Fig. 2-14). There are multiple etiologies of this systemic disease, including post-menopausal, age-associated, glucocorticoid-induced, secondary to cancer, androgen ablation, and aromatase inhibitors (Kanis 2002). All forms result in reduced bone strength and increased fracture risk, accompanied by a high degree of morbidity and mortality.

Post-menopausal osteoporosis is the most common form of the disease and results from a decline in gonadal hormone secretion following menopause. Rapid loss of trabecular BMD and, to a lesser extent, cortical loss are common in this condition (Kanis 2002).

Diagnosis is made by comparing the BMD of a patient to that of a healthy 20–29-year-old adult of the same gender. Systemic BMD at least 2.5 standard deviations below the average, referred to as a T-score, is used by the World Health Organization (WHO) to define osteoporosis (WHO 1994). Osteopenia, a less severe form of the disease, is diagnosed when T-scores are between −1.0 and −2.5 (Fig. 2-15).

Osteopetrosis

Osteopetrosis is a group of related diseases in which there is a pronounced increase in BMD due to abnormal bone turnover, and in some ways this is the opposite of osteoporosis. These conditions are inherited and the mode of transmission varies from autosomal dominant to autosomal recessive. Increases in BMD in this patient population are due to a variety of defects in osteoclastic bone resorption. These include higher or lower osteoclast numbers, impaired differentiation, deficiencies in carbonic anhydrase, the ability to form a ruffled border, and alterations in signaling pathways (Stark & Savarirayan 2009). In most cases, it is the ability of the osteoclast to create an acidic environment in the lacunae to resorb bone

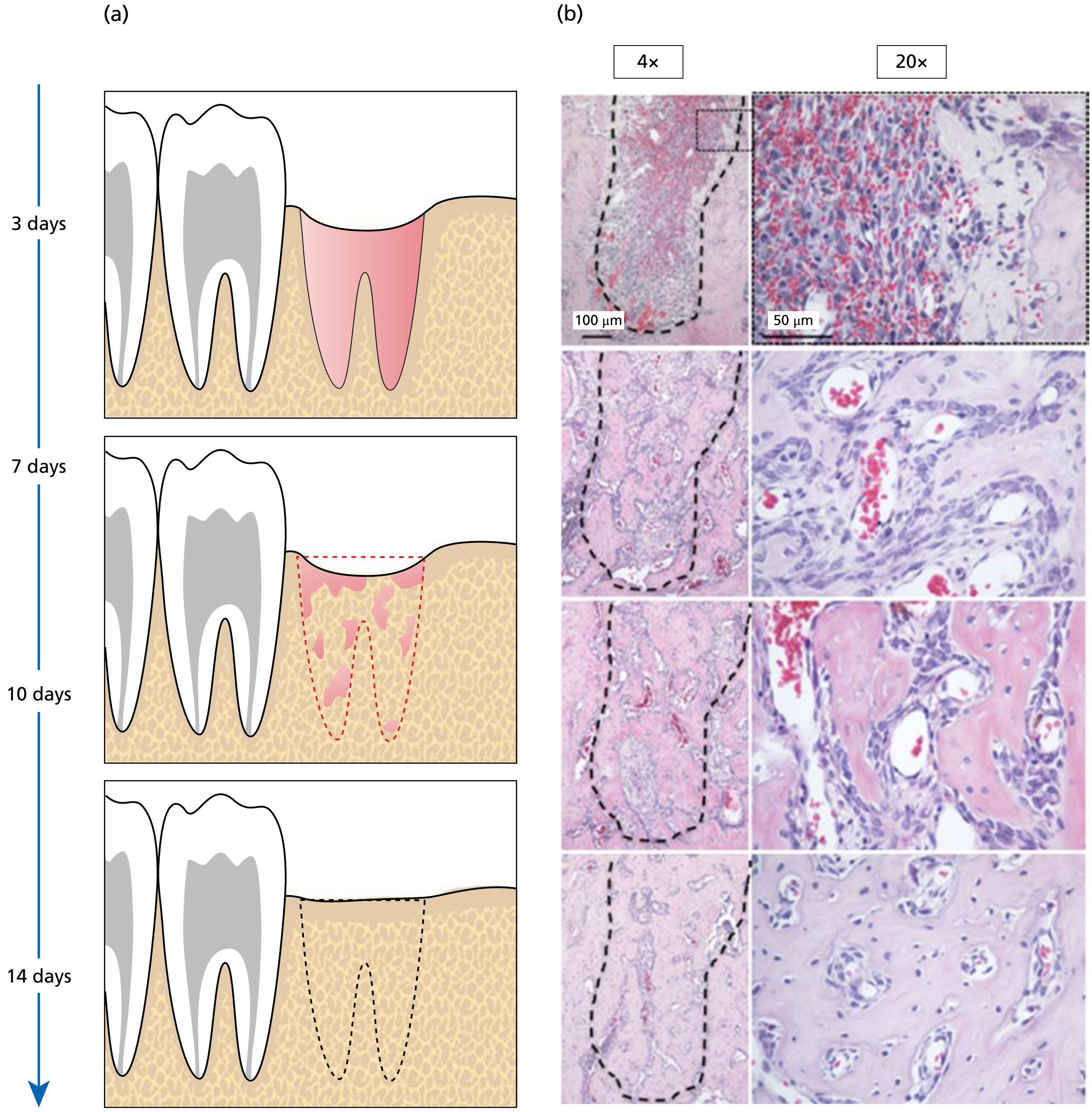


Fig. 2-12 Alveolar socket healing sites over time. (a) Rodent extraction model. Sequence of events that characterizes healing during the initial 14 days. (b) Hematoxylin and eosin (H&E) staining for tooth extraction site healing. The histologic images to the right of the healing area (black dashed lines) clearly capture the regeneration of the bone within the alveolar process. Note the clearly visible blood clot at day 3 (3d). At day 7 (7d), the cell density in the defect area is higher. At day 10 (10d), the defect site appears to be filled by a condensed mesenchymal tissue. Finally, by day 14 (14d), an integration of the newly formed bone to the original socket walls is noted.

that is in some way compromised, ultimately resulting in a net increase in bone formation (Fig. 2-16).

Osteomalacia

Vitamin D is essential for the metabolism of calcium and phosphorus in the body, which are the key minerals required for bone formation (Holick 2007). Vitamin D deficiency, or the inability to absorb the vitamin, is a common condition, especially in Northern climates since vitamin D is obtained primarily through sunlight exposure and diet. Other conditions may also predispose to vitamin D deficiency, such as oncogenic or benign tumors and liver disease.

When inadequate vitamin D is available, mineralization of the bones is impaired, resulting in a condition referred to as osteomalacia. When the disease occurs in children, it is referred to as rickets. The key features of osteomalacia are bones that contain a normal collagen matrix and osteoid structure, but lack proper mineralization, resulting in the softening of bones (Russell 2010). Osteomalacia differs from osteoporosis in that osteomalacia alters bone as it is developing, whereas osteoporosis weakens bones that have already formed (Fig. 2-17).

Severity ranges widely from an asymptomatic presentation to death in early childhood. Despite the increase in bone density, the newly formed bone is of

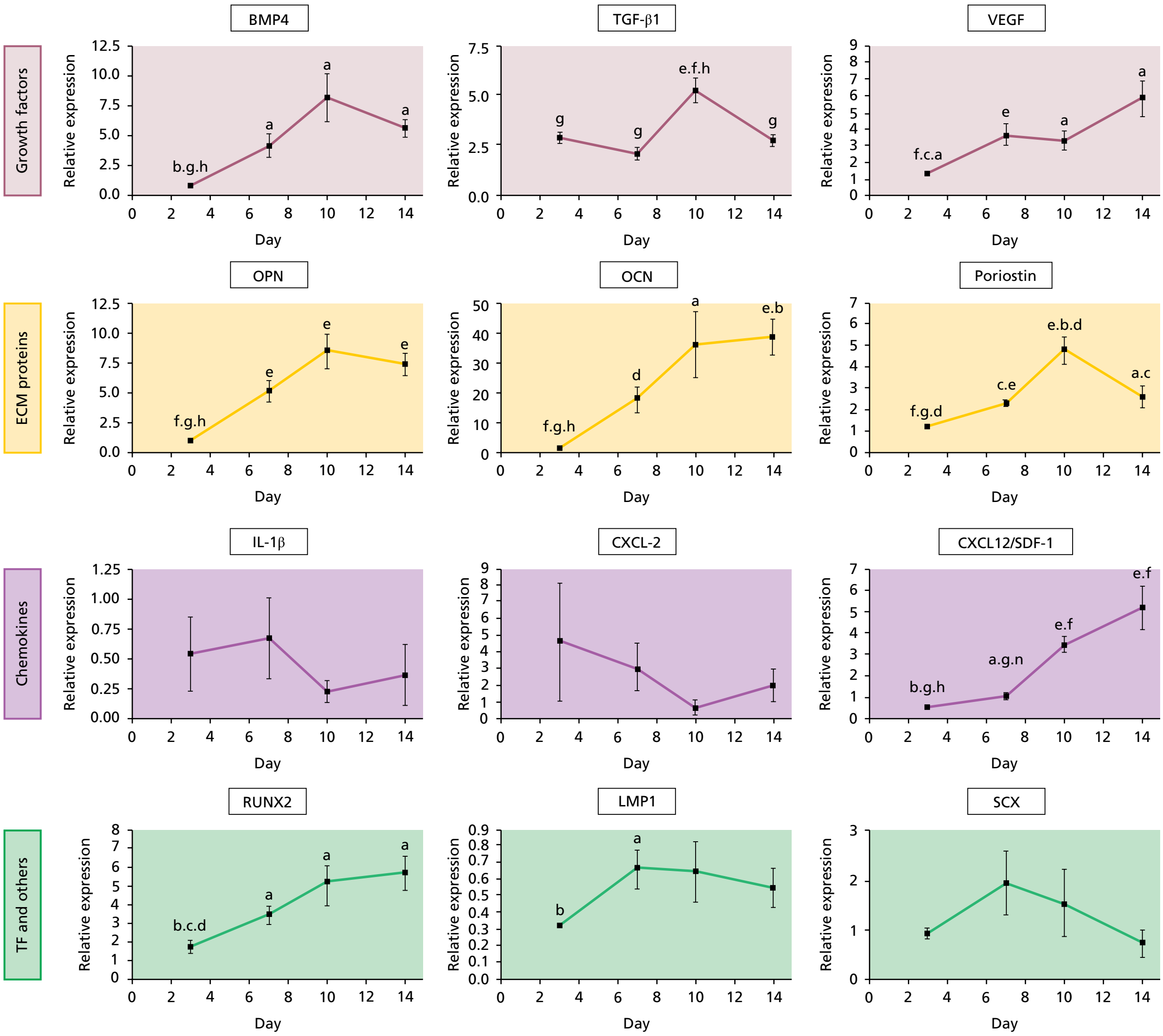


Fig. 2-13 Gene expression pattern of tooth extraction healing sites. Laser capture microdissection (LCM) analysis of genes associated with wound healing categorized them into three different groups: those for growth factors/chemokines, extracellular matrix proteins (ECM), and transcription factors (TF). The evaluation of gene expression over time captured the molecular dynamics that drive the bone healing process. Three expression patterns were evident. (1) Genes whose expression was slowly increased during the healing process: those for growth factors (*BMP4*, *BMP7*, *Wnt10b*, and *VEGF*), transcription factors (*RUNX2*), and extracellular matrix proteins related to mineralized tissue (*OPN* and *OCN*) were in this group; very interestingly, CXCL12 (SDF-1) gradually increased during extraction socket healing. Transforming growth factor beta 1 (TGF-β1) increased at the mid-stage of healing (day 10) and then decreased, and periostin (POSTN), a target gene of TGF-β1, had the same expression pattern. (2) Genes that were highly expressed at early time points and were down-regulated at later stages. Genes for chemokines IL-1β, CXCL2, and CXCL5 belonged to this category, although no statistical difference was seen due to the limited number of animals analyzed. Expression of *Wnt5a* and *Wnt4* also seemed to decrease during healing. (3) Genes that were constitutively expressed. LIM domain mineralization protein (*LMP-1*) and tendon-specific transcription factor *SCX* were in this group.

poor quality and symptoms include increased fracture incidence, neuropathy, and short stature. Treatment of osteomalacia involves reversing the vitamin D deficiency status, usually through dietary supplementation combined with removing the cause of the deficiency. Early management of this condition may involve a bone marrow transplant.

Osteonecrosis

When ischemia occurs in bone for an extended period of time, often due to an interruption in blood supply, cell death occurs. Cells from a hematopoetic lineage are most prone to the negative effects of ischemia and cannot survive for >12 hours without an adequate blood supply (Steinberg 1991). Cells directly responsible for bone mineralization and turnover – osteoblasts, osteoclasts, and osteocytes – are less prone to anoxia, although cell death occurs in these cells after 48 hours of anoxia. If the blood supply resumes quickly, healing may occur and the bone may recover. However, after this critical time period passes, the bone in question will necrose, requiring partial or total resection, followed by reconstruction.

Osteonecrosis has multiple etiologies including radiation, bisphosphonate use, steroid use, hypertension, and in some cases arthritis or lupus. Bisphosphonate-related osteonecrosis of the jaw (ONJ) is of growing concern in the dental field. ONJ is defined as an area of exposed bone that does not heal within 8 weeks after identification by

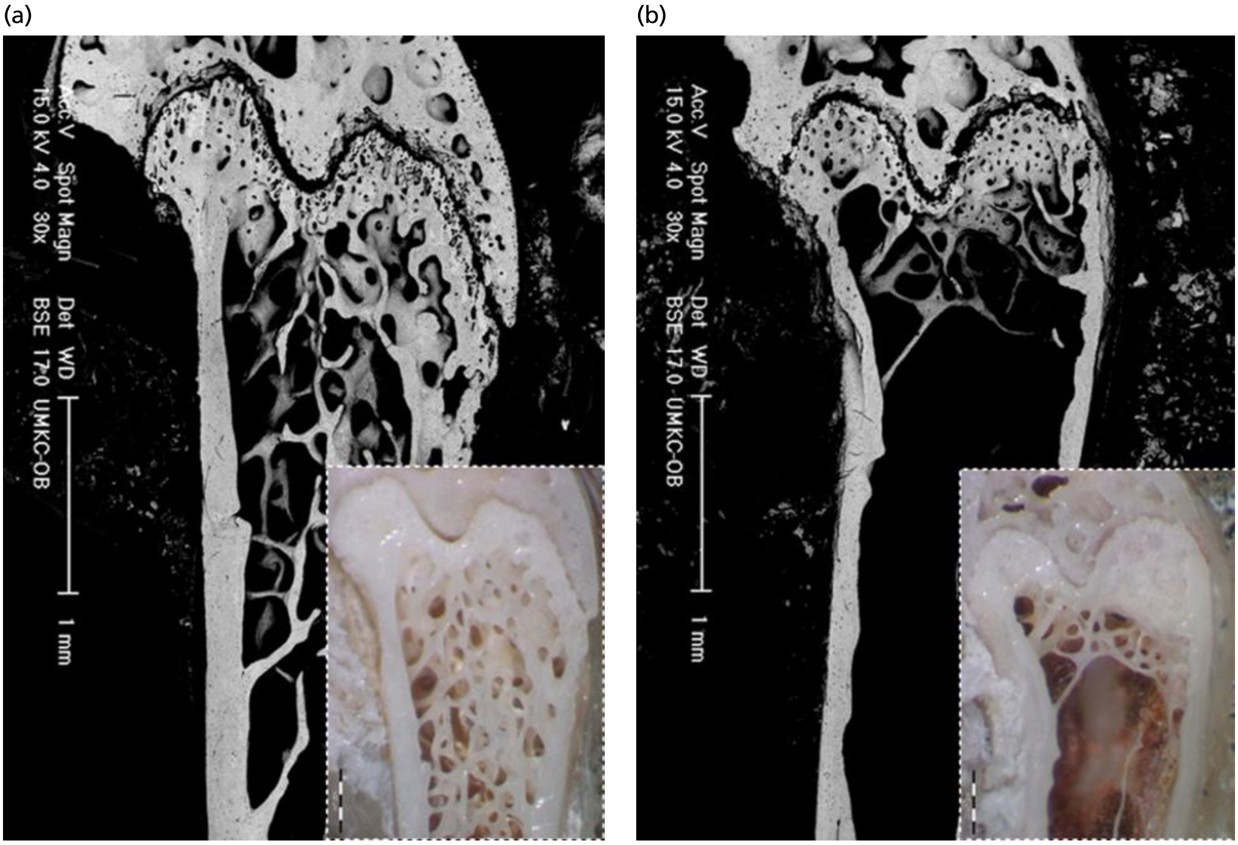
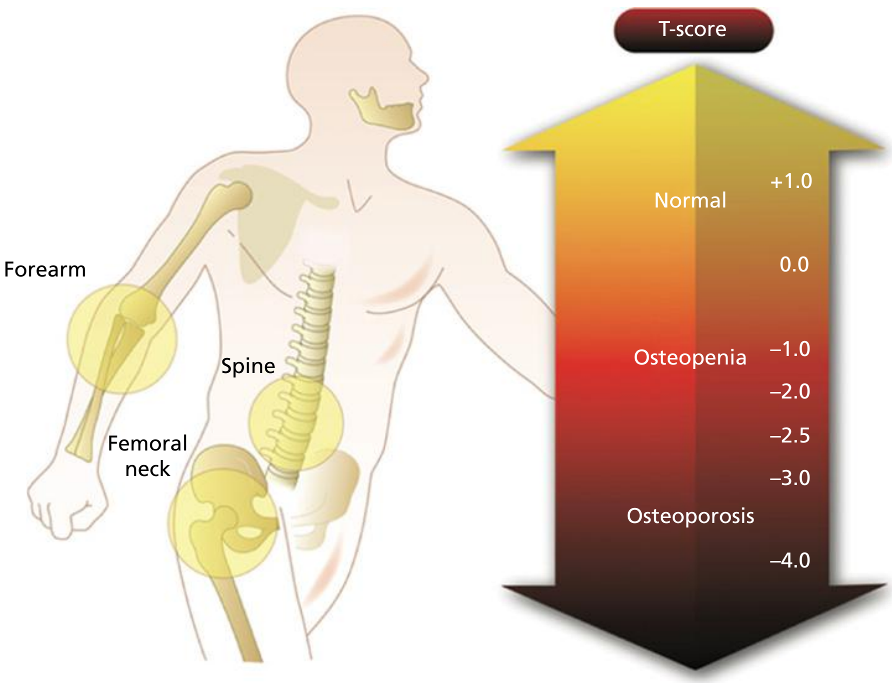


Fig. 2-14 Osteoporosis. In contrast to the image in (a), that in (b) illustrates microscopic bone morphologic changes associated with osteoporosis, such as decreased cortical thickness, in addition to a marked decrease in trabecular number and connectivity. As this process continues over time, there is further deterioration of the internal architecture with a significant impact on the ability of the bone to sustain compressive forces without failure.



$$\text{T-Score} = \frac{\text{Measured BMD} - \text{young adult mean BMD}}{\text{Young adult standard deviation}}$$

Fig. 2-15 Bone mineral density (BMD). Dual-energy X-ray absorptiometry (DEXA) is considered the preferred technique for measurement of BMD. The sites most often used for DEXA measurement of BMD are the spine, femoral neck, and forearm. The World Health Organization defines osteoporosis based on "T-scores". T-scores refer to the number of standard deviations above or below the mean for a healthy 30-year-old adult of the same sex as the patient.

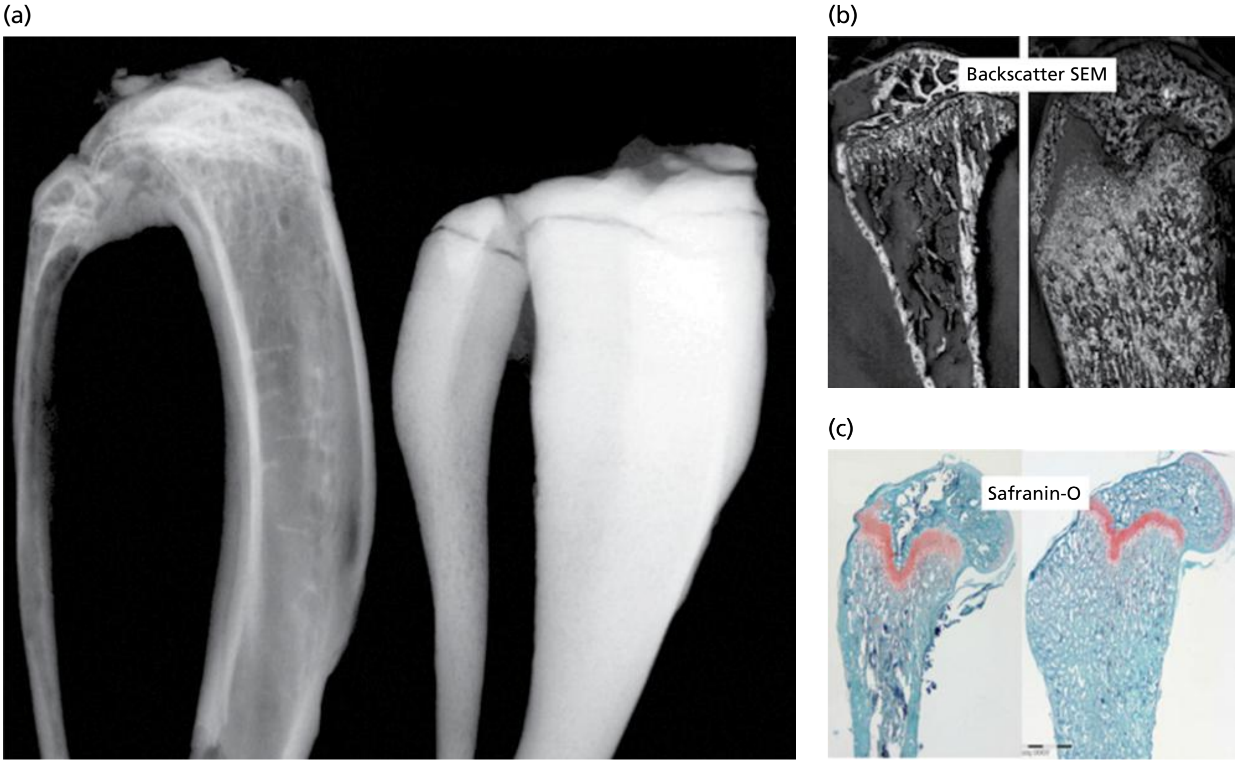


Fig. 2-16 Osteopetrosis. Increased density and deposits of mineralized bone matrix are a common finding in those with osteopetrosis. (a) Obliteration of the bone marrow cavity; (b) backscatter SEM; (c) staining with safranin-O.

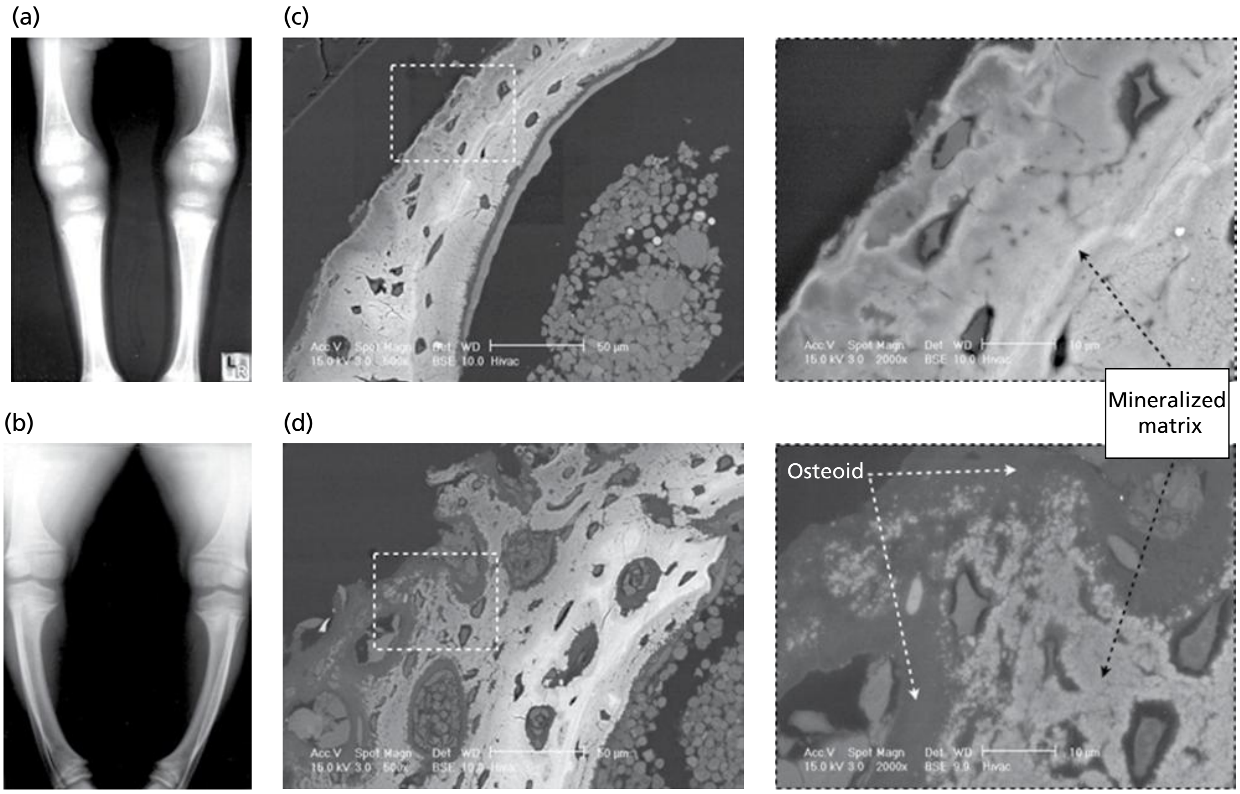


Fig. 2-17 Osteomalacia. (a, c) Normal matrix mineralization and maturation. (b, d) In osteomalacia, large hypomineralized zones accompanied by an increase in osteoid/immature matrix deposits are present.

a healthcare provider (Khosla *et al*. 2008). Patients diagnosed with bisphosphonate-related ONJ cannot have had prior radiation to the craniofacial regions. Oral bisphosphonate use is associated with lower risk and an incidence of 0.01–0.04%, compared to patients taking intravenous bisphosphonates who have a higher incidence of ONJ at 0.8–12% (Vescovi & Nammour 2011). This is likely due to the higher dosing regimen and disease being treated. Oral bisphosphonates are typically used to treat osteoporosis, whereas intravenous bisphosphonates are given for the treatment of Paget's disease, multiple myeloma, and other conditions.

Osteomyelitis

Osteomyelitis is an infection of the bone and can be classified according to the source of infection, prognosis, bone anatomy, host factors, and clinical presentation (Calhoun & Manring 2005). Open fractures, surgery, and conditions such as diabetes mellitus and peripheral vascular disease increase the risk of developing osteomyelitis. Osteomyelitis from a hematogenous source is much more common in the pediatric population.

A definitive diagnosis of osteomyelitis is made by isolation of the bacteria in conjunction with diagnostic imaging, but can be challenging. Treatment involves antibiotic therapy in conjunction with drainage, debridement, and other appropriate surgical management, including bone stabilization and skin grafting (Conterno & da Silva Filho 2009).

Osteogenesis imperfecta

Osteogenesis imperfecta (OI) is a group of genetic disorders of impaired collagen formation leading to decreased bone quality. Fractures, bone fragility, and osteopenia are common features of the disease. OI is relatively rare, with an incidence of 1 in 10 000 births. Autosomal dominant and recessive forms exist, although the autosomal dominant form is more common (Michou & Brown 2011).

The clinical presentation of OI has features in common with other diseases of bone metabolism, including fractures, bone deformities, and joint laxity. In addition, distinct features of OI include hearing loss, vascular fragility, blue sclerae, and dentinogenesis imperfecta. Type I collagen defects, including interruptions in interactions between collagen and non-collagenous proteins, weakened matrix, defective cell–cell and cell–matrix relationships, and defective tissue mineralization contribute to the etiology of the autosomal dominant form (Forlino *et al*. 2011). In the recessive form, deficiency of any of the three components of the collagen prolyl 3-hydroxylation complex results in a reduced ability of type I procollagen to undergo post-translational modification or folding. The severity of the disease, as well as the presence of defining features, varies widely.

Multiple therapeutic options are employed to treat the symptoms of OI, including surgery, collaboration with hearing, dental, and pulmonary specialists, and medication such as bisphosphonates and recombinant human growth hormone.

Other disorders

Several other conditions can affect bone homeostasis, including primary and secondary hyperparathyroidism, Paget's disease, and fibrous dysplasia.

Hyperparathyroidism is an overproduction of PTH, which promotes resorption of calcium and phosphorus from bone to increase serum calcium to normal levels (Unnanuntana *et al*. 2011). Primary hyperparathyroidism is most commonly caused by a parathyroid gland adenoma, whereas secondary hyperparathyroidism occurs when PTH production is overstimulated in response to low serum calcium. Hyperparathyroidism often presents with no symptoms and is discovered at routine screenings. The clinical presentation is very similar to that of rickets. Treatment includes identifying and eliminating the initiating cause.

Paget's disease is a condition where bone metabolism is significantly higher than normal, with boneformation exceeding that of resorption (Noor & Shoback 2000). This results in excessive bone formation and may affect one or multiple bones. The pelvic bone is most commonly affected. The affected bones, despite having increased bone formation, are weak and deformed. This is due to irregular collagen fiber formation within the bones. Bisphosphonate therapy is effective at decreasing bone turnover in this patient population, although this carries with it an increased risk of developing ONJ. Approximately 0.01–0.04% of patients taking bisphosphonates for the treatment of Paget's disease develop ONJ (Vescovi & Nammour 2011).

Fibrous dysplasia may affect multiple bones, but in 60% of cases, only one bone is affected (Michou & Brown 2011). It most commonly presents in childhood. Fibrous dysplasia lesions form in the medullary cavity extending to the cortical bone and are comprised of hyaline cartilage, immature woven bone, and osteoblast progenitor cells. Symptoms of this condition include fractures and bone pain. Notably, this condition has other craniofacial symptoms, including craniofacial bone deformities, exophthalmos, and dental abnormalities.

Conclusion

It can be appreciated that the dynamic nature of bone and its associated structures serves as an important organ system that supports form and function of the skeleton. This chapter serves to demonstrate the complexity of the developmental process of dental and craniofacial bone formation and homeostasis during health and disease.

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References

Ammann, P. & Rizzoli, R. (2003). Bone strength and its determinants. *Osteoporosis International* **14 Suppl** 3, S13–18.

Bonewald, L.F. (2002). Osteocytes: A proposed multifunctional bone cell. *Journal of Musculoskelet and Neuronal Interactions* **2**, 239–241.

Bonewald, L.F. (2007). Osteocytes as dynamic multifunctional cells. *Annals of the New York Academy of Science* **1116**, 281–290.

Bonewald, L.F. & Johnson, M.L. (2008). Osteocytes, mechanosensing and wnt signaling. *Bone* **42**: 606–615.

Burger, E.H., Klein-Nulend, J., van der Plas, A. & Nijweide, P.J. (1995). Function of osteocytes in bone –their role in mechanotransduction. *Journal of Nutrition* **125**, 2020S–2023S.

Burr, D.B., Martin, R.B., Schaffler, M.B. & Radin, E.L. (1985). Bone remodeling in response to in vivo fatigue microdamage. *Journal of Biomechanics* **18**, 189–200.

Calhoun, J.H. & Manring, M.M. (2005). Adult osteomyelitis. *Infectious Diseases Clinics of North America* **19**, 765–786.

Conterno, L.O. & da Silva Filho, C.R. (2009). Antibiotics for treating chronic osteomyelitis in adults. *Cochrane Database of Systematic Reviews* **3**, CD004439.

Fazzalari, N.L. (2011). Bone fracture and bone fracture repair. *Osteoporos International* **22**, 2003–2006.

Forlino, A., Cabral, W.A., Barnes, A.M. & Marini, J.C. (2011). New perspectives on osteogenesis imperfecta. *Nature Reviews Endocrinology* **7**, 540–557.

Hadjidakis, D.J. & Androulakis, I.I. (2006). Bone remodeling. *Annals of the New York Academy of Science* **1092**, 385–396.

Harkness, L.S. & Bonny, A.E. (2005). Calcium and vitamin D status in the adolescent: Key roles for bone, body weight, glucose tolerance, and estrogen biosynthesis. *Journal of Pediatric and Adolescent Gynecolgy* **18**, 305–311.

Holick, M.F. (2007). Vitamin D deficiency. *New England Journal of Medicine* **357**, 266–281.

Kanis, J.A. (2002). Diagnosis of osteoporosis and assessment of fracture risk. *Lancet* **359**, 1929–1936.

Khosla, S., Burr, D., Cauley, J. *et al.* (2008). Oral bisphosphonate-induced osteonecrosis: Risk factors, prediction of risk using serum CTX testing, prevention, and treatment. *Journal of Oral and Maxillofacial Surgery* **66**, 1320–1321; author reply 1321–1322.

Knothe Tate, M.L., Adamson, J.R., Tami, A.E. & Bauer, T.W. (2004). The osteocyte. *International Journal of Biochemistry and Cell Biology* 36, 1–8.

Lerner, U.H. (2006). Inflammation-induced bone remodeling in periodontal disease and the influence of post-menopausal osteoporosis. *Journal of Dental Research* **85**, 596–607.

Lin, Z., Rios, H.F., Volk, S.L., Sugai, J.V., Jin, Q. & Giannobile, W.V. (2011). Gene expression dynamics during bone healing and osseointegration. *Journal of Periodontology* **82**, 1007–1017.

Lips, P., Hosking, D., Lippuner, K. *et al.* (2006) The prevalence of vitamin D inadequacy amongst women with osteoporosis: An international epidemiological investigation. *Journal of Internal Medicine* **260**, 245–254.

Ma, Y.L., Dai, R.C., Sheng, Z.F. *et al*. (2008). Quantitative associations between osteocyte density and biomechanics, microcrack and microstructure in ovx rats vertebral trabeculae. *Journal of Biomechanics* **41**: 1324–1332.

Marotti, G. (2000). The osteocyte as a wiring transmission system. *Journal of Musculoskeletal and Neuronal Interactions* **1**, 133–136.

Marotti, G. & Palumbo, C. (2007). The mechanism of transduction of mechanical strains into biological signals at the bone cellular level. *European Journal of Histochemistry* **51 Suppl** 1, 15–19.

McCauley, L.K. & Nohutcu, R.M. (2002). Mediators of periodontal osseous destruction and remodeling: Principles and implications for diagnosis and therapy. *Journal of Periodontology* **73**, 1377–1391.

Michou, L. & Brown, J.P. (2011). Genetics of bone diseases: Paget's disease, fibrous dysplasia, osteopetrosis, and osteogenesis imperfecta. *Joint Bone Spine* **78**, 252–258.

Noor, M. & Shoback, D. (2000). Paget's disease of bone: Diagnosis and treatment update. *Current Rheumatology Reports* **2**, 67–73.

Parfitt, A.M. (1995). Bone remodeling, normal and abnormal: A biological basis for the understanding of cancer-related bone disease and its treatment. *Canadian Journal of Oncology* **5 Suppl** 1, 1–10.

Parfitt, A.M. (2002). Life history of osteocytes: Relationship to bone age, bone remodeling, and bone fragility. *Journal of Musculoskeletal and Neuronal Interactions* **2**: 499–500.

Raisz, L.G. (2005). Clinical practice. Screening for osteoporosis. *New England Journal of Medicine* **353**, 164–171.

Ranly, D.M. (2000) Craniofacial growth. *Dental Clinics of North America* **44**: 457–470, v.

Rodan, G.A. (1992). Introduction to bone biology. *Bone* **13 Suppl** 1: S3–6.

Russell, L.A. (2010). Osteoporosis and osteomalacia. *Rheumatic Diseases Clinics of North America* **36**, 665–680.

Schepetkin, I. (1997). Osteoclastic bone resorption: Normal and pathological. *Annals of the New York Academy of Science* **832**, 170–193.

Sims, N.A. & Gooi, J.H. (2008). Bone remodeling: Multiple cellular interactions required for coupling of bone formation and resorption. *Seminars in Cell Developmental Biology* **19**, 444–451.

Stark, Z. & Savarirayan, R. (2009). Osteopetrosis. *Orphanet Journal of Rare Diseases* **4**, 5.

Steinberg, M.E. (1991). Osteonecrosis of the hip: Summary and conclusions. *Seminars in Arthroplasty* **2**, 241–249.

Unnanuntana, A., Rebolledo, B.J., Khair, M.M., DiCarlo, E.F. & Lane, J.M. (2011). Diseases affecting bone quality: Beyond osteoporosis. *Clinical Orthopaedics and Related Research* **469**: 2194-2206.

Vaananen, H.K. & Laitala-Leinonen, T. (2008). Osteoclast lineage and function. *Archives of Biochemistry and Biophysics* **473**, 132–138.

Vescovi, P. & Nammour, S. (2011). Bisphosphonate-related osteonecrosis of the jaw (BRONJ) therapy. A critical review. *Minerva Stomatology* **59**, 181–203, 204–113.

WHO (1994). Assessment of fracture risk and its application to screening for postmenopausal osteoporosis. Report of a WHO study group. *World Health Organization Technical Report Series* **843**, 1–129.